Bradley L. Davis

Computer Engineer

me@bradleydavis.tech 206-484-7570 bradleydavis.tech github.com/WattsUp

Professional Experience

Schweitzer Engineering Laboratories // Pullman, WA

Lead Hardware Engineer // January 2025 - Present

- Developed and maintained SEL product lifecycle analysis tools using Flask, HTMX, Python, and SQL-backed data models. Aggregated incongruent engineering and manufacturing datasets into workflows used by teams representing 20% of the company.
- Partnered with R&D and in-house manufacturing stakeholders to turn process gaps into internal software workflows. Improved visibility into product status, manufacturability concerns, and lifecycle decision points.
- Built full-stack tooling for 10 Gigabit Ethernet and PoE type testing, including hardware fixtures, microcontroller measurement, and web-based setup support. Streamlined regulatory pre-test and post-test workflows.
- Initiated an SEL-owned GNSS-disciplined rackmount clock prototype outside normal duties. Built across PCB design, FPGA RTL, Linux drivers, C++ daemons, Yocto, Flask/HTMX UI, and Prometheus telemetry.

Hardware Engineer // July 2022 - December 2024

- Led development of GNSS-disciplined clock enhancement to add new timing signals with nanosecond accuracy. Collaborated directly with the customer to understand requirements and prototyping circuits in KiCAD.
- Mentored & supervised engineering interns.
- Automated test equipment using Python and VISA to reproducibly analyze units for functional performance.

Associate Hardware Engineer // July 2021 - June 2022

- Qualified electronic components that meet or exceed electrical and regulatory equivalence.
- Utilized oscilloscopes, digital multimeters, and network analyzers to functionally test hardware.
- Evaluated PCB layout in ANSYS SIwave simulations and recommended alterations that improved signal integrity and EMC.

Hardware Engineering Intern // May 2018 - June 2021

- Supported rugged industrial computer development and UL, IEC, and IEEE regulatory type testing.
- Developed C++ Ethernet reliability software and a manufacturing test tool that reduced defects.

Education

Colorado State University

Jan 2026 - Dec 2026 (expected)

M.Eng. Computer Engineering

Washington State University

Aug 2017 - Dec 2021

B.S. Electrical Engineering

Magna Cum Laude

Math & Physics minors

Relevant Skills

Software

Python, Flask, HTMX, HATEOAS, SQLAlchemy, SQLite, pytest, uv, Chart.js, JSON APIs

Data & Tools

Data modeling, dashboarding, workflow automation, reporting, internal tool development

Engineering Context

Product lifecycle analysis, manufacturing processes, regulatory testing, hardware validation

Collaboration

Requirements gathering, workflow translation, R&D/manufacturing process improvement